(3) $f(x)=\begin{cases}\frac{1}{4} & \text{for} \quad x=1,3\\ \frac{1}{16} & \text{for} \quad x=0,4\\ \frac{3}{8} & \text{for} \quad x=2\end{cases}$, $F(x)=\begin{cases}0 & \text{for} \quad x<0\\ \frac{1}{16} & \text{for} \quad 0\le x<1\\ \frac{5}{16} & \text{for} \quad 1\le x<2\\ \frac{11}{16} & \text{for} \quad 2\le x<3\\ \frac{15}{16} & \text{for} \quad 3\le x<4\\ 1 & \text{for} \quad 4\le x<\infty\end{cases}$

(4) (i) 8 (ii) $F(x)=\begin{cases}0 & \text{for} \quad x<0\\ \frac{1}{8} & \text{for} \quad 0\le x<1\\ \frac{3}{8} & \text{for} \quad 1\le x<2\\ 1 & \text{for} \quad 2\le x<\infty\end{cases}$ (iii) $\frac{7}{8}$

(5) (i)

x	-1	0	1	2	3
$f(x)$	0.15	0.20	0.25	0.25	0.15

(ii) $P(X<1)=0.35$ (iii) $P(X\ge 2)=0.40$

(6)(i) $\frac{1}{6}$ (ii) $\frac{17}{36}$ (iii) $\frac{5}{6}$ (7) (a)

x	0	1	2	3	4
$f(x)$	$\frac{1}{2}$	$\frac{1}{10}$	$\frac{1}{5}$	$\frac{1}{10}$	$\frac{1}{10}$

(b) $\frac{4}{5}$ (c) $\frac{2}{5}$

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(1) 4 (2) (i)0.16 (ii) 0.3 (iii) 0.75 (3) (i) $\frac{1}{400}$ (ii) $F(x)=\begin{cases}0, & x<200\\ \frac{x}{400}-\frac{1}{2}, & 200\le x\le 600\\ 1 & x>600\end{cases}$ (iii) $\frac{1}{2}$

(4) (i) $\frac{1}{3}$ (ii) $1-e^{-\frac{x}{3}}$ (iii) $1-e^{-1}$ (iv) $e^{-\frac{5}{3}}$ (v) $1-e^{-\frac{4}{3}}$

(5) (i) $F(x)=\begin{cases}0 & x\le -1\\ \frac{x^2}{2}+x+\frac{1}{2} & -1\le x<0\\ -\frac{x^2}{2}+x+\frac{1}{2} & 0\le x<1\\ 1 & 1\le x\end{cases}$ (ii) 0.75

(6) (i) $f(x)=F'(x)=\begin{cases}0 & x<0\\ \frac{1}{2}(2x+1) & 0\le x<1\\ 0 & 1\le x\end{cases}$ (ii) 0.285

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